Demonstration of Proposed Features

Date:	21 October 2026	**Team ID:**	Piyush-Balodiya-GenAI-2026
Project Name:	Multimodal Construction Intelligence Platform (CONSTRUCTAI)	**Maximum Marks:**	1 Mark

Demonstration of Proposed Features (Part 1 - Core Vision & Diagnostics):

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S. No	Feature Name	Description	Status	Demonstrated (Yes / No)	Remarks & Output
1	**Multimodal PPE Detection**	Real-time vision recognition of missing helmets, safety vests, and harnesses from CCTV & drone site imagery.	Implemented	**Yes**	Sub-second Groq LPU inference with visual bounding box overlays.
2	**IS 456 Structural Crack Diagnostics**	Categorizes concrete crack severity (hairline vs structural shear) against Indian Standard IS 456 limits.	Implemented	**Yes**	Generates instant chemical epoxy/grouting remediation steps.
3	**Predictive Weather Risk Advisory**	Syncs live Doppler rain radar & wind thresholds to prevent concrete pour washouts and crane accidents.	Implemented	**Yes**	Accurate Open-Meteo API rainfall risk forecasting verified.

Demonstration of Proposed Features

Demonstration of Proposed Features (Part 2 - BOQ, PDF & Alarms):

Tracking completion and verification of material estimation, compliance reporting, and perimeter alerting modules.

S. No	Feature Name	Description	Status	Demonstrated (Yes / No)	Remarks & Output
4	**Automated BOQ Material Estimation**	Calculates dynamic quantity takeoffs for cement, aggregates, and rebar steel from site blueprints.	Implemented	**Yes**	Instant material volume & cost calculation displayed on portal.
5	**Executive PDF Audit Generator**	Generates downloadable, tamper-proof EHS safety logs and structural repair blueprints.	Implemented	**Yes**	Automated ReportLab PDF generation with zero server data retention.
6	**Perimeter Hazard Alarms & Sirens**	Triggers multimodal visual red banners and audible chime alerts for danger zone violations.	Implemented	**Yes**	Sub-second alarm playback on crane lifting and pit zones.

Feature Implementation Summary:

Overall verification metrics confirming end-to-end execution across all planned platform deliverables.

Metric	Count / Value
Total Features Proposed	**6 Features**
Total Features Implemented	**6 Features**
Total Features Demonstrated	**6 Features**
Overall Implementation Rate (%)	**100%**